take time and result in significant overhead in Tx confirmation time compared with that of a centralized system. The transactions execute sequentially. The full nodes store the entire chain. Thus, transaction rates are not satisfactory compared with those of centralized applications, which affects scalability. In this section, let’s examine some solutions that address scalability.

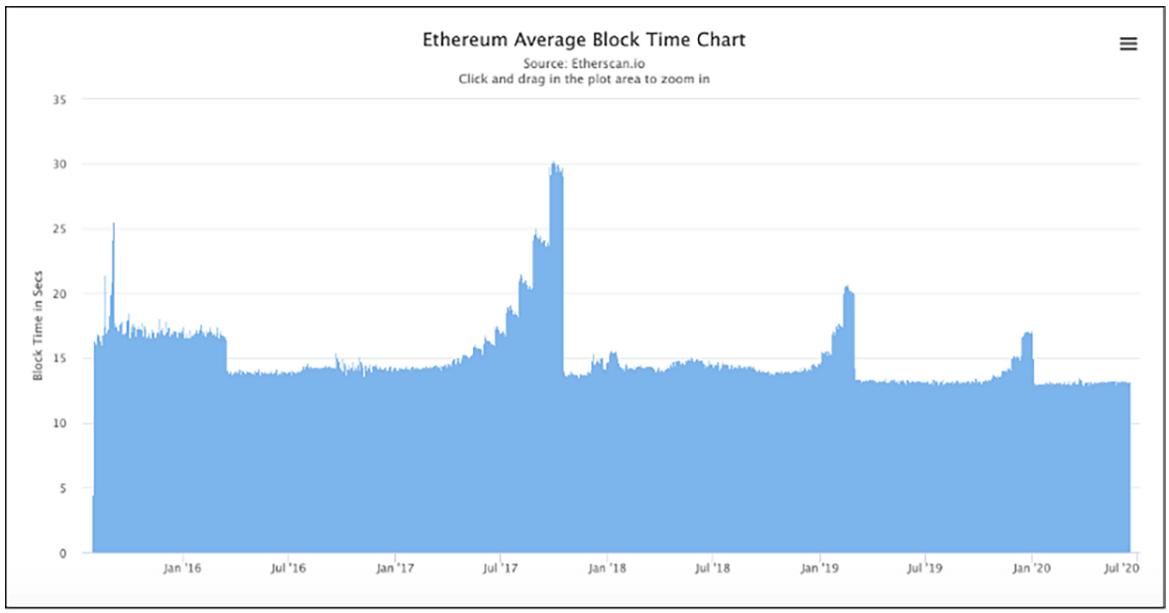


Figure 12.4 Ethereum average block time chart

Figure 12.4 shows average transaction time from etherscan.io from January 2016 to July 2020, the average block time, ranging from 12 seconds in 2020 to a maximum 30 seconds in 2017. A credit card transaction can be confirmed in less than a second, whereas it takes an average of 10 seconds for Ethereum. It is expected that in the latest version of Ethereum, transactions per second will improve to 3,000. But the Visa credit card network is capable of handling a load of 65,000 transactions per second. Thus, scalability is an area that requires a lot of attention, and this is an opportunity for you to contribute creative solutions.